WORKED SOLUTION

STEP II 2012, Question 12

Probability and Statistics • Conditional Probability

STEP II 2012, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q12

The key idea

Pressing a switch *toggles* it, so the pool's hallway switch ends up on either because it started on and was left alone, or because it started off and was pressed. The kitchen half is a red herring in disguise: a switch that is on with probability $\frac{1}{2}$ stays on with probability $\frac{1}{2}$ whether or not it is toggled. Once part (i) gives a single conditional probability, the seven Mondays are a binomial experiment, and "most likely value is 3" becomes a pair of inequalities between neighbouring binomial probabilities.

Solution

Part (i)

Define the events

- H: the guest presses the pool's switch in the hallway, so $\mathrm{P}(H) = \frac{1}{4}$;
- A: the hallway switches are all on at the start, so $\mathrm{P}(A) = p$;
- L: the pool light is on at the end.

The guest's choice of switch is made at random, so H is independent of A.

Hallway. After the press, the pool's hallway switch is on exactly when either A happens and H does not (it was on and untouched), or A fails and H happens (it was off and got toggled on). So

$$\mathrm{P}(\text{hallway pool switch on at end}) = p \cdot \tfrac{3}{4} + (1-p) \cdot \tfrac{1}{4} = \frac{1+2p}{4}.$$

Kitchen. The pool's kitchen switch starts on with probability $\frac{1}{2}$. If it is not pressed it stays as it was; if it is pressed it flips, and a switch that is off with probability $\frac{1}{2}$ becomes on with probability $\frac{1}{2}$. Either way it ends on with probability $\frac{1}{2}$, whatever the guest does in the kitchen.

The kitchen switches and the guest's kitchen choice are independent of everything in the hallway, so

$$\mathrm{P}(L) = \frac{1+2p}{4} \cdot \frac{1}{2} = \frac{1+2p}{8}.$$

For the conditional probability, $H \cap L$ requires the guest to press the pool's hallway switch, that switch to have started off (so that pressing turns it on), and the kitchen switch to end on:

$$\mathrm{P}(H \cap L) = \frac{1}{4} \cdot (1-p) \cdot \frac{1}{2} = \frac{1-p}{8}.$$

Hence

$$\mathrm{P}(H \mid L) = \frac{(1-p)/8}{(1+2p)/8} = \frac{1-p}{1+2p},$$

as required.

Part (ii)

Write $q = \dfrac{1-p}{1+2p}$. On each Monday we are given that the light was on, so by part (i) the pool's hallway switch was pressed with probability q, independently from week to week (the guests act independently and the starting states are fresh each Monday). So the number X of days on which it was pressed satisfies

$$X \sim \mathrm{B}(7, q).$$

We need $1-q$ as well:

$$1 - q = \frac{(1+2p) - (1-p)}{1+2p} = \frac{3p}{1+2p}, \qquad \text{so} \qquad \frac{q}{1-q} = \frac{1-p}{3p}.$$

(Here $p > 0$; if $p = 0$ then $q = 1$ and $X = 7$ with certainty, so 3 could not be the most likely value.)

Ask yourself How do we turn "3 is the most likely value" into something we can calculate?

Compare neighbouring probabilities. For $1 \le k \le 7$,

$$\frac{\mathrm{P}(X=k)}{\mathrm{P}(X=k-1)} = \frac{\binom{7}{k} q^k (1-q)^{7-k}}{\binom{7}{k-1} q^{k-1} (1-q)^{8-k}} = \frac{8-k}{k} \cdot \frac{q}{1-q}.$$

The factor $\frac{8-k}{k}$ strictly decreases as k increases, so the probabilities rise while this ratio exceeds 1 and fall once it drops below 1. Thus 3 is the (unique) most likely value exactly when the probabilities rise from $k=2$ to $k=3$ and fall from $k=3$ to $k=4$:

$$\frac{\mathrm{P}(X=3)}{\mathrm{P}(X=2)} = \frac{5}{3} \cdot \frac{q}{1-q} > 1 \qquad \text{and} \qquad \frac{\mathrm{P}(X=4)}{\mathrm{P}(X=3)} = \frac{4}{4} \cdot \frac{q}{1-q} < 1.$$

(These two conditions alone are enough for the direction we need: if 3 is most likely then certainly $\mathrm{P}(X=3) > \mathrm{P}(X=2)$ and $\mathrm{P}(X=3) > \mathrm{P}(X=4)$.)

Substituting $\frac{q}{1-q} = \frac{1-p}{3p}$ and multiplying through by the positive quantity $3p$:

- the second condition gives $1 - p < 3p$, that is, $p > \frac{1}{4}$;
- the first gives $\frac{5}{3}(1 - p) > 3p$, that is, $5 - 5p > 9p$, so $p < \frac{5}{14}$.

Hence $\frac{1}{4} < p < \frac{5}{14}$, as required.

Checks

- At $p = 0$ the hallway switches are all off, so the light can only be on if the pool switch was pressed: part (i) gives $\mathrm{P}(H \mid L) = 1$, as it should. At $p = 1$ the light can only be on if that switch was *not* pressed, and the formula gives 0.
- The endpoints correspond to $q = \frac{1}{2}$ (at $p = \frac{1}{4}$) and $q = \frac{3}{8}$ (at $p = \frac{5}{14}$). At $q = \frac{1}{2}$, $\mathrm{B}(7, \frac{1}{2})$ is symmetric, so 3 and 4 tie; at $q = \frac{3}{8}$, $(n+1)q = 3$ and 2 and 3 tie. So the strict inequalities are exactly right.
- We checked part (i) by enumerating every starting state and every choice of switch exactly with fractions, and part (ii) by computing the $\mathrm{B}(7, q)$ probabilities for p on a grid of step $\frac{1}{2800}$: the unique mode is 3 precisely for $\frac{1}{4} < p < \frac{5}{14}$.

Common slips

- Forgetting that pressing a switch that is already on turns it *off*: the hallway calculation needs both routes to "on".
- Computing the kitchen by cases when the answer is simply $\frac{1}{2}$; the case analysis is harmless but an easy place for arithmetic errors.
- Using p instead of q as the binomial parameter in part (ii): the conditioning on the light being on is what changes the probability of pressing the switch from $\frac{1}{4}$ to q.

Beyond STEP

Part (i) is Bayes' theorem in action: an observation (the light is on) updates the probability that the guest pressed the switch from $\frac{1}{4}$ to $\frac{1-p}{1+2p}$. Part (ii) runs the idea in reverse, asking which values of an unknown parameter p are consistent with a given most likely outcome, which is the seed of statistical inference and maximum likelihood estimation. The ratio-of-consecutive-terms trick for locating a mode works for many discrete distributions (Poisson, hypergeometric). Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and these distributions thoroughly.

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